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Conservation of Endangered Plants in Urban Fire-Prone Habitats

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Abstract. A number of endangered plant species occur in National Parks and associated bushland that are adjacent to urban areas in the Sydney Region of south-eastern Australia. Within these areas fire is a repeated and frequent disturbance event that potentially threatens not only the endangered flora but life and property as well. These threats were well illustrated by the impact of the fires that occurred in January 1994 in Sydney.

For the management of such fire-prone areas there is a need to identify those fire regimes likely to drive the endangered flora to extinction. Management of fire-sensitive, threatened *Darwinia* (Myrtaceae) and *Grevillea* (Proteaceae) species illustrates how conservation goals can be applied along with protection of life and property. A regime of frequent fire will lead to local population declines and extinction in both *Darwinia* and *Grevillea* species. Consequently, frequent burning on a regular basis for hazard reduction to protect property assets is inappropriate for the conservation of these species. *Darwinia* species also require fires that cause significant soil heating (high consumption of fine fuel) to break seed dormancy in soil stored seed, while *Grevillea* species require fires of a large spatial extent to minimise the impact of mammalian seed predators on seed outputs from the first few seed-crops produced by plants after a fire.

Keywords: Fire regime; Threatened species; *Darwinia*; *Grevillea*

Introduction

In the Sydney Region of southeastern Australia, much of the vegetation is highly flammable and fires are a common and frequent event. For example, during January of 1994, some 90% and 65% respectively of the two major, large urban reserves (Royal and Ku-ring-gai Chase National Parks) were burnt. In this rich and diverse floristic area, around 2000 plant species regularly experience fire. As a consequence, the management of fire is one of the principal challenges facing managers of natural ecosystems in the Sydney Region. Managers must attempt to protect life and property from the dangers posed by fire in urban bushland as well as conserving this biodiversity.

Given the high potential for extreme fire events, where effective control of wildfire is not always possible (as happened in January 1994), there is increasing pressure for widespread fuel reduction burning in urban bushland adjacent to human assets such as houses. Such areas contain a number of endangered, vulnerable or rare plant species that are of high conservation significance, as well as more common species, all of which will have particular fire requirements which may or may not be compatible with burning for fuel reduction purposes. Hence, there is a potential for a conflict over protection versus conservation priorities (Bradstock, Keith & Auld 1995, Bradstock & Scott 1995). To attempt to resolve such conflicts we must have a sound knowledge of the effects of different fire regimes on plant and animal species and as land managers apply an informed choice of appropriate fire regimes to ensure the maintenance of populations of plants and animals.

This paper presents data on the effect of different components of the fire regime on several threatened plant species in order to provide land managers with potential conservation prescriptions for threatened plant species. In particular, we identify those fire regimes that may cause populations to decline. Such prescriptions can then be incorporated into planning and the implementation of a fuel reduction program, so that conservation issues are not ignored and fire regimes which threaten plant conservation are avoided.

Study Area

The Sydney area in south-eastern Australia is a large urban area fringed with large bushland reserves and containing many smaller reserves (Figure 1). Within this area, the vegetation is dominated by sclerophyll heathlands, woodlands and forests, situated on infertile sandy soils derived from Hawkesbury sandstone (Beadle 1981), in a landscape of dissected sandstone plateaux. Rainfall is around 1200mm per annum with a peak in late summer. Average monthly maximum/minimum temperatures for

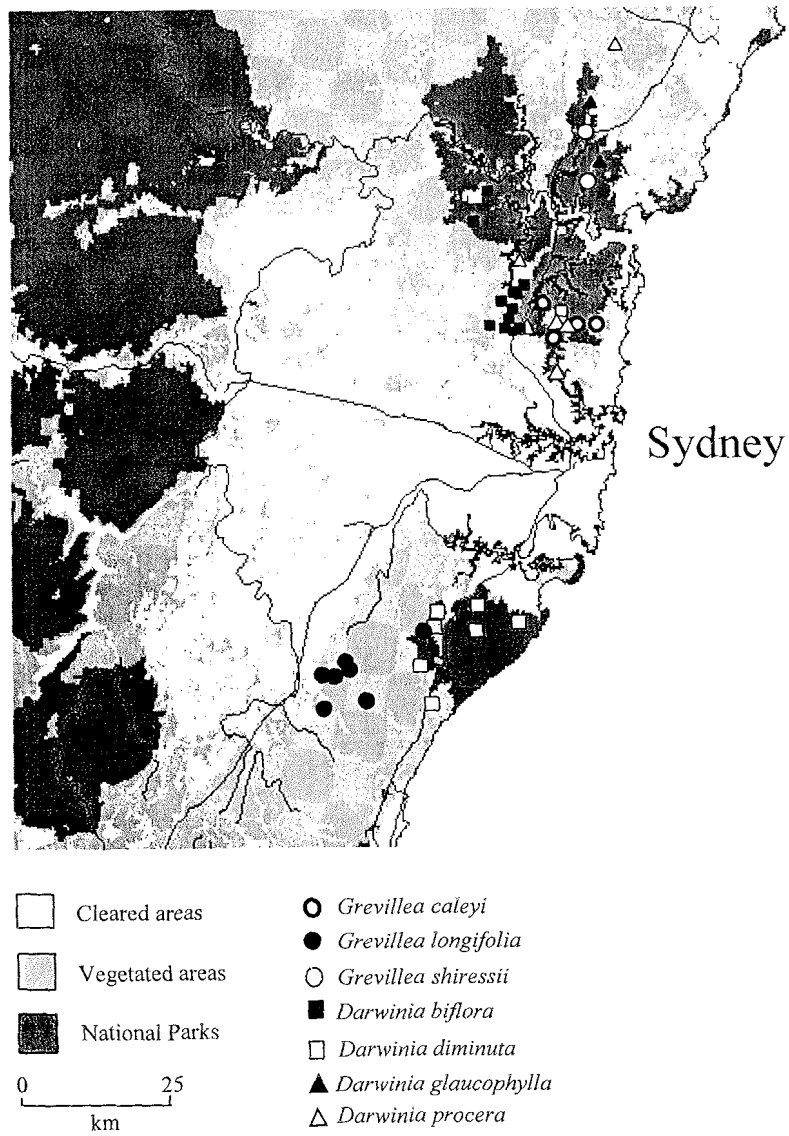


Figure 1. The Sydney urban area and adjacent bushland reserves showing the distribution of *Darwinia* and *Grevillea* study species.

Sydney are 26/18°C in summer and 16/8°C in winter. A broad range of fire regimes are possible in the Sydney Region. High intensity wildfires occur in the order of once every 5-10 years, while deliberate prescriptions of low intensity fires are commonly used for fuel reduction. Arson is a major source of wildfire ignitions.

The Fire Regime

For the long term survival of plants and animals we need to know how they are affected by the fire regime, ie. (frequency, intensity, season, spatial extent). Here we concentrate on fire-sensitive plants.

Fire frequency

All fires burn areas of vegetation that have been burnt some time before. Consequently, the vegetation is really a mosaic of fire histories, with some parts having had more fires than others over a given time period. We need to try and understand how important this unequal patchiness of fires through time is for the survival of plants.

Plants that are killed by fire rely on seed germination after fires to survive at a particular location. For these fire-sensitive plants, there must be sufficient time between successive fires for seedlings to mature and produce more seeds and hence, add seeds to the seedbank. In the Sydney area, this time will vary between species and loca-

tions (Benson 1985), from species that flower within 1-2 years after fire (*Actinotus helianthi, Acacia suaveolens*) to those that may take 4-8 years to reach maturity (*Banksia ericifolia*). If another fire should occur before these plants have matured, dramatic changes in the vegetation may occur. Fire-sensitive plants with seed stored in woody capsules usually release all their seeds after a fire and no residual seeds remain in cones. These plants are the group of species most susceptible to two fires in short succession. Plants that have seeds stored in the soil usually have some residual seedbank remaining in the soil following post-fire germination. For these species, a second fire before seedlings mature may severely reduce population numbers but it may not eliminate species locally. However, in some places, after some fires, the entire soil seedbank may be exhausted, eg. Restionaceae and Epacridaceae (Meney et al. 1994), *Grevillea* (Auld unpubl.) and local extinction will occur if two fires occur close enough together to kill seedlings from the first fire before they are mature. Ideally, as land managers with concerns for the conservation of biodiversity, we should allow the interval between fires to be long enough not only simply for plant species to mature, but at a minimum, several fruiting seasons beyond this so that a sufficiently large seedbank can be established before the next fire occurs.

Fire Intensity

How hot a fire is will depend on how much combustible fuel is available, how dry this fuel is, the topography of the area and the weather conditions, such as humidity and windspeed. When an area is burnt, the important thing for plants and animals is how much heat is produced, and in many areas around Sydney, this usually varies markedly even on a very small scale (i.e., over tens of meters). How much heat a fire produces can affect the survival of plants. In intense fires there is likely to be higher levels of resprouting plants killed, and longer periods needed before the next fire for recovery, than in mild fires. For plants, the more heat produced by a fire, the greater are the opportunities for seedling recruitment and the greater the magnitude of seedlings that are recruited (Auld & O'Connell 1991). Many plant species have seeds which lie dormant in the soil and for a number of these the amount of soil heating that occurs during the passage of a fire will govern the degree of germination after the fire. This heat response is widespread in many plant families, particularly legumes, in fire-prone communities throughout the world (Martin et al. 1975, Jeffery et al. 1988, Auld & O'Connell 1991, Keeley 1991). Bradstock and Auld (1995) have shown that in Hawkesbury sandstone soil around Sydney, it is the amount of fine fuel that is consumed during a fire that governs the degree of soil heating.

Fire season

Given the prevalence of arson and widespread hazard reduction burning for the protection of life and property, fires can now be expected to occur at just about any time of the year in the Sydney Region. The question is, for the long-term survival of plants does it matter when the vegetation is burnt? In Sydney, Bradstock and Bedward (1992) have suggested that seedling survival after fire is largely independent of season of burn because rainfall is very variable. Although there is a trend for wetter periods in late summer/autumn and drier periods in spring this pattern is not strong enough to allow accurate predictions about the impact of season of burn on seedling survival post-fire. Perhaps the most critical role season of burning may play is in influencing fire intensity. Fires in the hot dry summer months when fuel moisture is low will generally be of higher intensities than burns in winter when the vegetation is moister, an ambient temperatures are lower.

Fire Spatial Extent

What is the impact of the spatial extent of fires on plants? In Sydney, fires come in all shapes and sizes. Some burn very extensive areas, while others may only burn a hectare or so. Within the boundaries of a burnt area, fires may be widespread and burn most of the vegetation or leave large parts of the vegetation unburnt or simply lightly scorched. All fires will be different, depending on the terrain, rate of fire spread, amount of fuel available and its moisture content and weather conditions and natural or man-made fire breaks. This results in a patchiness of fires in area and intensity.

For plants, the spatial extent of fires may be important where burns occur on a very small scale (a few hectares). In such cases, native animals from surrounding unburnt vegetation may graze heavily on the new green shoots that appear after fire and hence, cause a decline in some plant species. These native animals may also consume many of the seeds that are produced in the first few years after fire for those fire-sensitive plants that are trying to replenish their soil seedbanks. This will slow the rate of recovery of such plants.

Study Species

The threatened species examined, *Darwinia* spp. (Myrtaceae) and *Grevillea* spp. (Proteaceae), are all killed by fire (fire-sensitive, see Gill 1981) and rely on germination from a soil seedbank for persistence at a site after a fire. They are all restricted to the vicinity of Sydney (Figure 1). Threat codes are from Briggs and Leigh (1996).

Darwinia biflora (Cheel) B.Briggs. Vulnerable. A small shrub to 1 m in height in woodland communities mainly in the Hornsby Plateau area in the northern suburbs of Sydney and associated with the Hammondville soil group (Burkitt, Wilson & Benson 1984). The largest populations occur in Ku-ring-gai Chase National Park, with small outliers in Lane Cove National Park, Marramarra National Park and Pennant Hills Park. This species is threatened by habitat loss, disturbance leading to weed invasion and an inappropriate fire regime.

Darwinia diminuta B Briggs. Rare. A small shrub to 60 cm in height in woodland and heath communities. Found on sandstone in a few localities in Royal National Park, Heathcote National Park, Garawarra State Recreation Area and Kelly's Falls to the south of Sydney and Garigal National Park in the northern Sydney suburbs. *D. diminuta* is threatened by habitat loss, site disturbance and an inappropriate fire regime.

Darwinia glaucophylla B. Briggs. Rare. A spreading prostrate shrub in heath. Confined to a few localities on sandstone in the Gosford area north of Sydney. *D. glaucophylla* is threatened by urban development and an inappropriate fire regime.

Darwinia procera B. Briggs. Rare. An erect shrub to 2 m in height in heath to open forest communities. Mostly confined to sheltered gullies and creek banks on sandstone. Restricted to a few localities between Garigal NP and Brisbane Water NP, in the northern suburbs of Sydney. *D. procera* is threatened by habitat loss, urban runoff and an inappropriate fire regime.

Grevillea caleyi R.Br. Endangered. A shrub to 4m by 4m occurring on laterite soils in open forests or low open forests and restricted to roughly a 6 x 6 km area centred on the northern Sydney suburb of Terrey Hills. Habitat destruction has divided the original distribution of the species into a series of discrete population fragments (Scott et al. 1995) separated by urban development (houses, roads, business parks etc). Some 85% of the former habitat of *G. caleyi* has been lost to urban development. Only two of the remaining sites are relatively undisturbed and only one of these two has any long-term protection as it occurs in Ku-ring-gai Chase National Park. The remaining large patches are highly disturbed. The major threats to the survival of *G. caleyi* arise from human interference in the environment. These threats range from loss of habitat through clearing, urbanisation and associated infrastructure, changes in landuse and an inappropriate fire regime, to localised disturbance at sites including invasion of weed species and the dumping of rubbish (Scott et al. 1995).

Grevillea longifolia R.Br. Rare. A bushy shrub 2-4 m in height by the same wide. It is found in the southern half of the Sydney Region and Woronora plateau and the Jamieson Valley of the Blue Mountains. Usually found in moist areas adjacent to creeks in open forest, but occasionally on escarpments and ridgetops in woodlands on Hawkesbury sandstone. *G. longifolia* is threatened by urban development, runoff, weed encroachment and an inappropriate fire regime.

Grevillea shiressii Blakley. Vulnerable. A bushy shrub 2-5 m in height. Found only on the banks of two creeks in closed forest in Brisbane Water National Park in the north of Sydney on soils derived from Narrabeen group geology. *G. shiressii* is threatened by disturbance (a fire trail passes through one population) and an inappropriate fire regime.

The Effects of the Fire Regime on Threatened Plant Species

For all study species, demographic data were collated on threatened *Darwinia* and *Grevillea* species (Scott 1986, Auld et al. 1991, Auld et al. 1993, Auld and Tozer 1995, Scott et al. 1995) and the impacts of components of the fire regime were assessed. A summary of the significant demographic characteristics of our threatened study species is presented in Table 1.

Darwinia spp. (Myrtaceae)

As the *Darwinia* spp. studied are all killed by fire, a fire-free period is needed after any one fire to allow seedlings to mature and replenish the soil seedbank. Plants can produce a few fruits two years after fire. As plants grow, fecundity increases and for all *Darwinia* species, soil seedbanks should increase rapidly 5-10 years after a fire. Most of the annually produced seed-crop is released in a viable, dormant state and would be incorporated into this seedbank, although there is some yearly variation in the proportion of non dormant seeds produced (eg. *D. glaucophylla*, Table 1).

For *D. biflora*, some 75% of the dormant seed released in any one annual seed-crop becomes non dormant and has germinated by 24 months after release, i.e., most seeds from any one seed-crop are short-lived in the soil. The seedbank should buildup in the soil rapidly after a site is burnt, but the seedbank will also decline rapidly should adult plants senesce. The low level of seed longevity in the soil in *D. biflora* (Table 1) suggests that both adult survival and the level of adult fecundity are the critical factors that govern persistence of the soil seedbank. As seeds decay rapidly in the soil, adult plants must regularly produce seeds to maintain a soil seedbank. The longevity of annual seed inputs to the soil seedbank is currently unknown for the other *Darwinia* spp. However, for *D. diminuta*, *D. glaucophylla* and *D. procera* the seedbank size is likely to be much larger than in *D. biflora*, given the higher fecundity levels in these species (Figure 2).

The fate of those seeds in the soil which lose their dormancy and germinate is unknown. It seems likely that

Table 1. Comparative demographic features of Threatened *Darwinia* and *Grevillea* species.

Characteristic	*D. biflora*	*D. diminuta*	*D. glaucophylla*	*D. procera*	*G. caleyi*	*G. longifolia*	*G. shiressii*
Primary juvenile period	1.5-2 years	2-3 years	unknown	2-3 years	2.5-5 years	3-5 years	unknown
Flowering	annually	annually	annually	annually	annually	annually	annually
Fecundity post-fire (see Figs 2 & 3)	low initially (all species have increasing fecundity with plant size)				low initially	low initially	low initially
Mortality rate[1]	0.17	0.012-0.216	0.037-0.393	0.027-0.046	0.22-0.5	0.002-0.073	unknown
Predispersal seed predation (% seeds lost)	6-31	37-52	<2	0-8	20-31	unknown	57
Postdispersal seed predation (% seeds lost)	0	0	0	0	82-92	92-93	60
Dispersal agent	ants	ants	ants	ants	none known	none known	ants
Seed viability (%)	74-99	52-80	85-94	73-88	69-95	88-95	80-99
Seed dormancy at release(%)	88-100	79-97	53-99	94-100	97-98	40-73	99-100
Seed longevity in the soil (half-life)[2]	0.9 years	unknown	unknown	unknown	7.6 years	unknown	unknown
Dormancy broken by heat	yes	yes	yes	yes	no	no	no
Post-fire seedling pulse	yes	yes	yes	yes	yes	yes	unknown

[1] of adult plants greater than 10 years old and assuming log linear decay.
[2] based on exponential decay rates of seeds in the soil.

seedlings which emerge during the inter-fire period do not establish because of competitive pressure from an established cover of shrubs, herbs and grasses (c.f. *Acacia* spp. Weiss 1984, Auld 1986). Alternatively, some limited on-going establishment could be possible if seedlings were able to tolerate the existing cover of shrubs, herbs and grasses.

All *Darwinia* spp. showed some breaking of seed dormancy in response to heating. This suggests that the degree of soil heating during the passage of a fire, itself controlled by available fuel loads (Bradstock & Auld 1995), will influence the number of seeds within the seedbank that germinate after a fire. Near the soil surface lethal soil temperatures (>120°C) are likely to cause some seed death. The seedbank of these *Darwinia* spp. will be unaffected in a fire where soil heating is below the critical level needed to break seed dormancy, i.e., fires consuming small quantities of fuel.

Grevillea spp. (Proteaceae)

All *Grevillea* species studied were killed by fire and relied on seed germination after a fire to maintain populations. For all species, the magnitude of the soil seedbank will be largely controlled by the levels of adult fecundity, mammal seed predation and adult mortality.

In *G. caleyi* and *G. longifolia*, seedlings do not flower and produce seeds before 2.5-5 years of age. Fecundity is low in *G. caleyi* with only about 3% of flowers resulting in mature seed (Scott 1986). The level of fecundity in both *G. caleyi* and *G. longifolia* increases until around 10 years of age in response to plant growth (Figure 3). No fecundity estimates have been made for *G. shiressii*. Seeds had a high level of viability and were largely released in a dormant state in *G. caleyi* and *G. shiressii*. In *G. longifolia*, a large fraction of seeds were released in a non dormant state (Table 1), although this is variable across populations and years. Predispersal seed predation is known to be high in *G. caleyi* and *G. shiressii* with significant losses to the weevil *Cydmaea dorsalis* Lea (Coleoptera: Curculionidae) in *G. caleyi* and to the wasp *Eurytoma* sp. (Hymenoptera: Eurytomidae) in *G. shiressii*. In *G. caleyi* and *G. longifolia*, seed dispersal is minimal. At maturity the fruits dehisce, dropping the seed to the ground beneath the parent plant. The seeds lack any lipid body attachment that would make them attractive to ants (Auld 1995). In contrast, *G. shiressii* seeds have a lipid body attachment and some 28% of seed are dispersed by ants. In all three threatened *Grevillea* spp. and other more common *Grevillea* spp., post-dispersal seed predation is very high (Auld 1995, Auld & Denham in prep.). Most of the seeds lying on the soil surface are consumed by

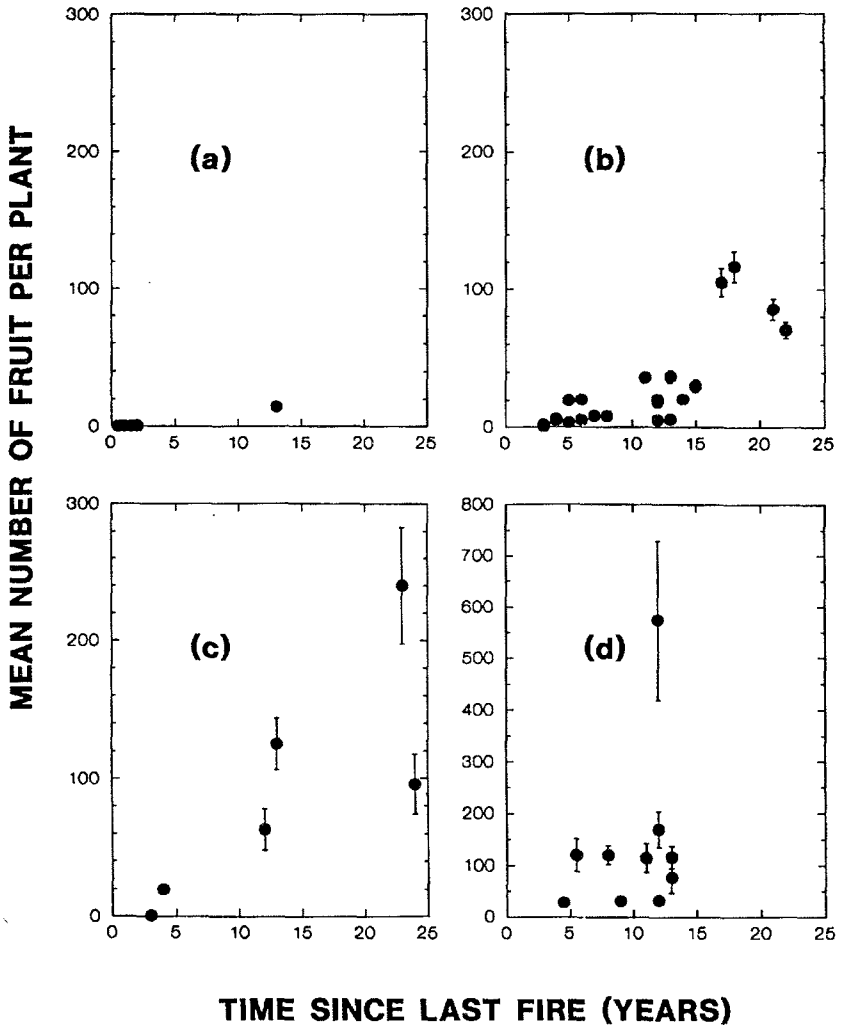


Figure 2. Fecundity after fire in *Darwinia*. a) *D. biflora*; b) *D. diminuta*; c) *D. glaucophylla;* and d) *D. procera*.

two native mammals, swamp wallabies (*Wallabia bicolor*) and bush rats (*Rattus fuscipes* and possibly other *Rattus* species). As a result, very few of the developing fruits from any one season contribute seeds to the seedbank. The relationship between mammal densities and the size of area burnt needs to be examined. In the past, some *G. caleyi* sites have been subjected to small patch burns. In these burnt areas, the level of seed losses to mammals was similar to adjacent unburnt areas and the establishment of a soil seedbank was expected to be retarded.

The major difference between *G. caleyi* and *G. longifolia* was the level of adult mortality within a population. Adult mortality was high in *G. caleyi*, while there was little adult mortality in *G. longifolia*. This means that as populations age beyond about 15 years post-fire, annual seed output (and the amount of seed successfully reaching the soil seedbank) will decline in *G. caleyi* at a much greater rate than in *G. longifolia*. The soil seedbank in *G. caleyi* is small and declines to low levels some 20 years after fire. In *G. longifolia*, the soil seedbank, although limited in magnitude by mammal seed predation, is more persistent through time. This suggests that *G. longifolia* will better tolerate long unburnt periods that *G. caleyi*. Comparable data is not available for *G. shiressii*.

In Sydney, heat from a fire does not break seed dormancy in *Grevillea* species, although a post-fire flush of seedlings is evident (Auld and Tozer 1995, Table 1). Hence, the amount of fuel consumed in a fire may have no effect on post-fire germination levels. Such germination should be governed by the response of seeds to another fire-related germination cue as well as the rate of decay of dormant seeds to a non dormant state in the soil seedbank. Leachates from charred wood (Keeley 1991), smoke (Brown 1993, Dixon et al. 1995) and nitrate and ammonium ions (Thanos and Rundel 1995) have all been shown to be fire-related cues that break seed dormancy

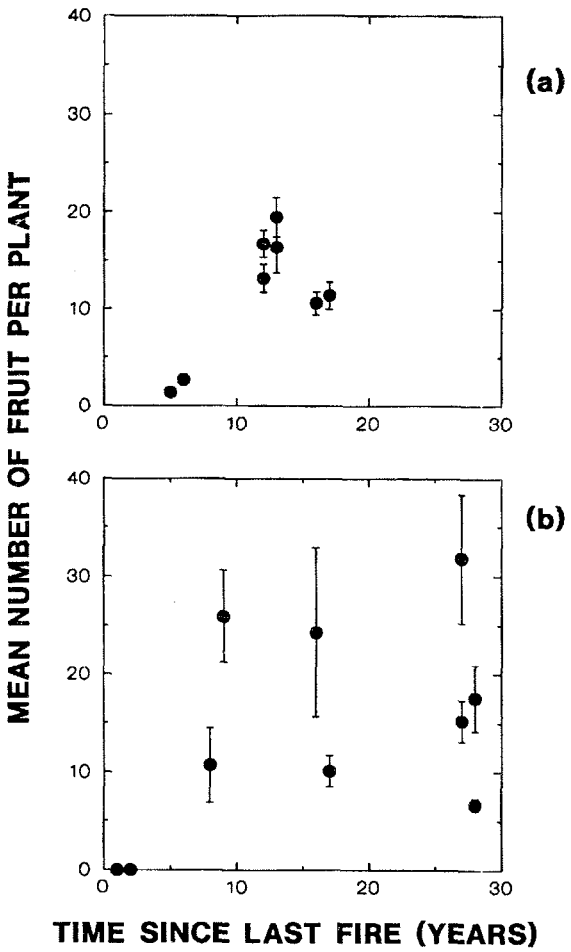


Figure 3. Fecundity after fire in *Grevillea*. a) *G. caleyi*; and b) *G. longifolia*.

in fire-prone habitats. Which cues are critical in *Grevillea* species in Sydney remains to be investigated.

The key management indicators for healthy populations of *G. caleyi* in long unburnt (>15-20 year old) areas would appear to be the levels of adult survival and mammal seed predation. Such areas now occur as very small isolated pockets within urban areas. At sites where survival is low and mammals are destroying most of the seeds produced in any one year, seeds in the soil will continue to persist, but the magnitude of the soil seedbank will decline and the population is likely to be reduced after the next fire. This suggests that consideration should be given to burning such sites before the seedbank declines too much. Any such burning should be of a large spatial extent to reduce the densities of mammalian seed predators after fire.

Implications for Fire Management

What are the implications for the management of these obligate seeding species? The most critical aspect of fire management for obligate seeders is the impact of frequent fires on these populations. Population declines or local extinctions are predicted after 2-3 fires occurring in close succession and hence, minimum fire-free intervals are needed to maintain the species. The fire free-intervals will vary depending on the rate of replenishment of the soil seedbank after any one fire and the proportion of seeds that emerge from the seedbank after a fire (ie, the degree to which the seedbank is depleted after a fire). Estimates of the minimum fire free intervals needed to maintain populations of threatened *Darwinia* and *Grevillea* species vary from 5-10 to 8-12 years (Table 2). It is not only important to maintain these threatened species, but also the species with which they co-habit. These suggested minimum fire-free intervals do not contradict with suggested minimum fire-free intervals for other plants species that co-habit with threatened *Darwinia* and *Grevillea* species (Bradstock and Auld 1987). Consequently, for the long-term survival of threatened species of *Darwinia* and *Grevillea* and their associated vegetation communities, under a regime of repeated fires, care should be taken to accurately record the fire histories of all remaining populations. This will allow sites potentially threatened by high fire frequency to be identified.

In *Darwinia* species, where heat is the fire-related cue that breaks seed dormancy, some residual seedbank should be present after any single fire. The amount and location of this seed will depend on the level of soil heating during the passage of a fire (Bradstock and Auld 1995). In *Grevillea*, the fire-related dormancy breaking cue is unknown, although it is not simply heat (Auld and Tozer 1995). In *Grevillea*, a high proportion of seeds within the soil seedbank may emerge after any one fire and in some

Table 2. Fire regime requirements for threatened *Darwinia* and *Grevillea* species. No effect of fire season is currently known.

Species	Fire-free Interval (years)		Required Fine fuel consumption	Spatial Extent of burnt area required
	Min.	Max.		
Darwinia biflora	5-10	20-30?	high	unknown
D. diminuta	5-10	20-30?	high	unknown
D. grandiflora	5-10	>40?	high	unknown
D. procera	5-10	>40?	high	unknown
Grevillea caleyi	10-12	15-25	unknown	large
G. longifolia	10-12	25-40?	unknown	large
G. shiressii	unknown	unknown	unknown	large

cases no residual seed may remain after a fire (Auld unpubl.). Hence, *Darwinia* species are likely to be more resilient than *Grevillea* species to two fires close enough together to prevent seedlings from the first fire from maturing. For *Grevillea* species, there is a significant risk of local extinctions under such a scenario, in the same manner as for many fire-sensitive species with a canopy seedbank where the entire seedbank is exhausted after a single fire (Lamont et al. 1991).

Where protection of life or property dictates a fire frequency less than that prescribed in Table 2, consideration should be given to alternative fuel reduction methods in sites containing threatened *Darwinia* or *Grevillea* species, eg. manual fuel removal. These areas will usually be those sites adjacent to urban areas and large expanses of natural vegetation. At other sites, where small patches of remnant vegetation are now isolated within the urban area, there is evidence that when populations of *G. caleyi* are left unburnt for 15-20 years or longer, the number of adult plants declines such that inputs to the soil seedbank are minimal. Consideration should also be given to burning such areas if all adults within the population have died and no seedling recruitment has occurred.

For *Darwinia* species, fires allowing high levels of ground fuel consumption should produce soil temperatures that cause seed dormancy to be broken. Fires where little ground fuel consumption occurs will not produce high enough soil temperatures to break seed dormancy and hence, are likely to lead to population declines, although some post-fire germination should also occur in response to seeds in the soil seedbank decaying from a dormant to a non dormant state through time. For *Grevillea* species, the role fire intensity (or its components) plays in controlling germination is currently unknown.

Finally, as post-dispersal seed predators can consume most of the annual seed-crops in threatened *Grevillea* species, consideration should be given to the size and patchiness of fires in areas containing these plants. Fires that burn large spatial areas are recommended to reduce the impact of mammalian seed predators in the first 10 years post-fire. If small scale burns are used to protect or manipulate sites containing threatened *Grevillea* species, then the mammalian seed predators may have a large impact on the amount of seed that reaches the soil seedbank. If large areas are burnt, there will be declines in the densities of the mammalian species (cf. Fox and McKay 1981) and less impact on the number of seeds reaching the soil seedbank from annual seed-crops, at least in the first few fruiting years after fire. The complex nature of this relationship requires further investigation. Where large area burns are not possible consideration should be given to fencing sites to exclude these seed predators while the seedbank is being replenished.

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